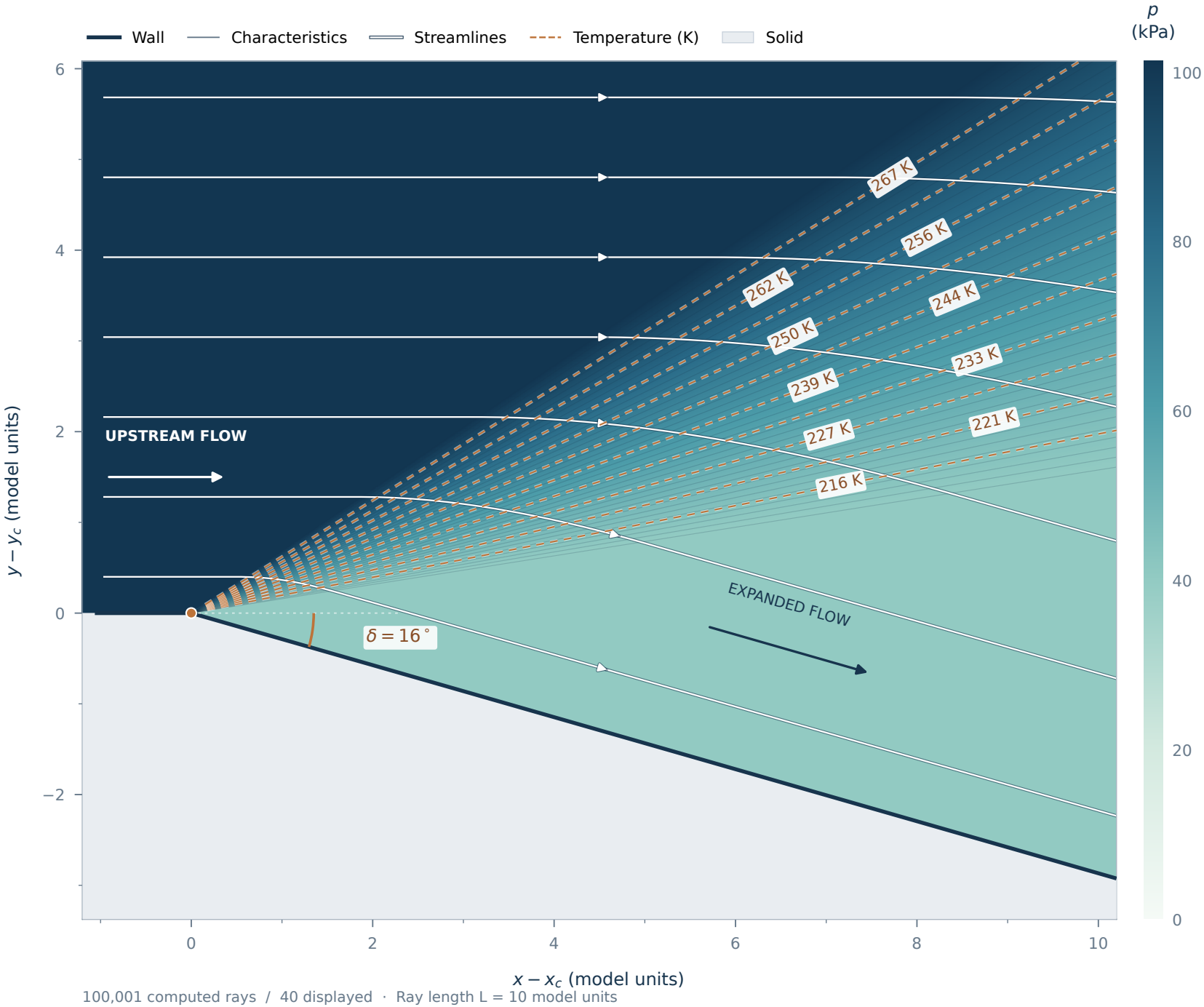
Prandtl–Meyer expansion

16° TURN

Static pressure field with temperature isotherms



100,001 computed rays / 40 displayed · Ray length L = 10 model units

p: absolute static pressure · T: static temperature · Field interpolated between computed rays

FLOW STATES

UPSTREAM

M_1 **1.775**

p_1 101.3 kPa

T_1 273.1 K

DOWNSTREAM

M_2 **2.369**

p_2 40.2 kPa

T_2 209.8 K

CHANGE THROUGH FAN

Mach	**+33.5%**
Pressure	**-60.3%**
Temperature	**-23.2%**

PLANAR / ISENTROPIC

Steady, inviscid, constant γ